



Investigação Operacional

Armazém de Paletes

2017/2018

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Introdução

No âmbito da unidade curricular de Investigação Operacional, foi-nos proposto a realização do projeto “O Armazém de Paletes” que consiste em otimizar a gestão de um armazém. Armazém este que usa paletes de dois tipos (industrial e europeia), para armazenar produtos, e as paletes são guardadas em alvéolos de *racks*. A otimização consiste em melhorar certos aspetos do armazém, como a disposição das paletes, reduzir a distância total percorrida pelos operadores para assim reduzir o custo associado à movimentação das paletes.

Este trabalho está dividido em três fases, sendo que as primeiras duas ignoram o parâmetro A (altura dos alvéolos das *racks*) e a última fase já tem em conta este parâmetro.

Fase I – Problema de Afetação

Nesta fase foi proposto que cada alvéolo contivesse apenas uma paleta, sendo esta a restrição para esta fase.

Tendo em conta isto e que é necessário armazenar 469 paletes (figura 1), são necessários 469 alvéolos.

Tipo	Altura	Qt		Tipo	Altura	Qt
1	14	54		1	14	2
2	20	76		2	20	4
3	22	2		3	30	4
4	23	2		4	40	3
5	27	6		5	42	2
6	28	6		6	60	8
7	30	11		7	90	24
8	40	51				47
9	42	1				
10	60	174				
11	64	1				
12	64,2	1				
13	80	16				
14	90	17				
15	96	2				
16	100	2				
		422				
				422	Paleta 1200x1000	
				47	Paleta 1200x800	
				469	Total	

Figura 1

Para dividir as paletes pelas *racks* sendo que cada *rack* tem o mesmo número de paletes optamos por dividir o armazém em 67 *racks*, tendo 7 níveis cada uma. Optamos por estes valores, pois 7 é o divisor mínimo de 469 (diferente de 1) e 67 é o quociente. Os valores possíveis (mantendo todas as *racks* com o mesmo número de alvéolos), são 1,7,67 e 469, sendo que achamos mais “realista” as 67 *racks* com 7 alvéolos cada.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
																16
																17
	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
																33
																34
																35
	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	
																51
																52
	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	

Figura 2 – Disposição do armazém na fase I.

E as paletes com o parâmetro W (utilização anual) mais elevado devem estar presentes numa *rack* mais próxima da porta, pois a distância percorrida pelos operadores vai ser menor, otimizando distâncias e tempos.

Função objetivo:

$$\min: \sum_{i=1}^n \sum_{j=1}^{67} \sum_{k=1}^7 d_{i,j,k}$$

- n = n.º total de paletes;
- i = n.º de paletes;
- j = n.º de estantes/*racks*;
- k = n.º de prateleiras/níveis;
- d = distância percorrida por um operador para aceder a palete

NEOS Server Version 5.0
Job# : 6106723
Password : yfDgbaiQ
User : None

Solver :
milp:CPLEX:AMPL
Start : 2018-06-06
12:43:53

End	: 2018-06-06	11753	'x[26,1,28]'	1	48156	'x[103,5,50]'	1
12:44:03		12632	'x[27,7,36]'	1	48474	'x[104,3,33]'	1
Host	: NEOS HTCondor	12813	'x[28,3,16]'	1	49145	'x[105,6,34]'	1
Pool		13483	'x[29,6,16]'	1	49480	'x[106,4,34]'	1
		13861	'x[30,4,59]'	1	50090	'x[107,6,41]'	1
Disclaimer:		14387	'x[31,5,49]'	1	50633	'x[108,7,48]'	1
		14881	'x[32,6,7]'	1	51006	'x[109,6,19]'	1
This information is		15052	'x[33,1,44]'	1	51326	'x[110,4,4]'	1
provided without any express		15615	'x[34,3,4]'	1	51621	'x[111,1,31]'	1
or		16007	'x[35,1,61]'	1	52249	'x[112,3,56]'	1
implied warranty. In		16656	'x[36,4,40]'	1	52560	'x[113,1,32]'	1
particular, there is no		17039	'x[37,3,21]'	1	53198	'x[114,3,67]'	1
warranty		17526	'x[38,3,39]'	1	53908	'x[115,7,40]'	1
of any kind concerning		17840	'x[39,1,18]'	1	54118	'x[116,3,49]'	1
the fitness of this		18372	'x[40,2,14]'	1	54478	'x[117,2,7]'	1
information for any		18867	'x[41,2,40]'	1	55304	'x[118,7,29]'	1
particular purpose.		19263	'x[42,1,34]'	1	55444	'x[119,2,35]'	1
*****		19938	'x[43,4,39]'	1	56279	'x[120,7,66]'	1
*****		20582	'x[44,7,13]'	1	56580	'x[121,5,32]'	1
*****		20844	'x[45,4,7]'	1	56827	'x[122,2,11]'	1
File exists		21414	'x[46,5,41]'	1	57437	'x[123,4,18]'	1
You are using the solver		21935	'x[47,6,26]'	1	57787	'x[124,2,33]'	1
cplexamp.		22170	'x[48,2,60]'	1	58245	'x[125,2,22]'	1
Checking ampl.mod for		22934	'x[49,7,20]'	1	58704	'x[126,2,12]'	1
cplex_options...		23031	'x[50,1,50]'	1	59413	'x[127,5,51]'	1
Checking ampl.com for		23722	'x[51,5,4]'	1	59589	'x[128,1,26]'	1
cplex_options...		24203	'x[52,5,16]'	1	60207	'x[129,3,41]'	1
Executing AMPL.		24532	'x[53,3,10]'	1	60561	'x[130,1,60]'	1
processing data.		25088	'x[54,4,30]'	1	61061	'x[131,2,24]'	1
processing commands.		25733	'x[55,7,5]'	1	61689	'x[132,4,49]'	1
Executing on prod-exec-		25926	'x[56,2,64]'	1	62282	'x[133,6,39]'	1
5.neos-server.org		26721	'x[57,7,55]'	1	62764	'x[134,6,52]'	1
		26753	'x[58,1,20]'	1	62909	'x[135,1,63]'	1
219961 variables, all binary		27396	'x[59,3,60]'	1	63321	'x[136,1,6]'	1
938 constraints, all linear;		27993	'x[60,5,54]'	1	63809	'x[137,1,25]'	1
439922 nonzeros		28276	'x[61,3,2]'	1	64370	'x[138,2,50]'	1
469 equality		28783	'x[62,3,40]'	1	65146	'x[139,7,22]'	1
constraints		29126	'x[63,1,48]'	1	65244	'x[140,1,53]'	1
469 inequality		29623	'x[64,2,9]'	1	65790	'x[141,2,63]'	1
constraints		30415	'x[65,6,64]'	1	66355	'x[142,4,25]'	1
1 linear objective; 219961		30924	'x[66,7,37]'	1	66911	'x[143,5,45]'	1
nonzeros.		31398	'x[67,7,42]'	1	67201	'x[144,2,67]'	1
		31541	'x[68,2,51]'	1	67884	'x[145,6,13]'	1
CPLEX 12.7.0.0: threads=4		31961	'x[69,2,2]'	1	68328	'x[146,5,55]'	1
CPLEX 12.7.0.0: optimal		32635	'x[70,5,6]'	1	68623	'x[147,3,15]'	1
integer solution; objective		32881	'x[71,1,51]'	1	69013	'x[148,2,3]'	1
67736		33671	'x[72,6,37]'	1	69667	'x[149,4,54]'	1
0 MIP simplex iterations		34182	'x[73,7,12]'	1	69928	'x[150,1,47]'	1
0 branch-and-bound nodes		34490	'x[74,4,52]'	1	70462	'x[151,2,45]'	1
No basis.		35010	'x[75,5,36]'	1	70967	'x[152,3,14]'	1
: _varname[j]		35587	'x[76,7,10]'	1	71342	'x[153,1,54]'	1
_var[j] :=		35844	'x[77,3,66]'	1	71934	'x[154,3,43]'	1
282 'x[1,5,14]'	1	36180	'x[78,1,67]'	1	72386	'x[155,3,26]'	1
639 'x[2,3,36]'	1	37002	'x[79,7,18]'	1	72879	'x[156,3,50]'	1
1405 'x[3,7,65]'	1	37407	'x[80,6,21]'	1	73265	'x[157,2,34]'	1
1869 'x[4,7,60]'	1	37814	'x[81,5,26]'	1	73901	'x[158,4,67]'	1
1949 'x[5,2,6]'	1	38362	'x[82,6,38]'	1	74387	'x[159,5,17]'	1
2557 'x[6,4,11]'	1	38786	'x[83,5,60]'	1	75004	'x[160,7,31]'	1
3230 'x[7,7,14]'	1	39138	'x[84,4,10]'	1	75209	'x[161,3,35]'	1
3294 'x[8,1,11]'	1	39733	'x[85,6,2]'	1	75738	'x[162,4,28]'	1
4097 'x[9,6,10]'	1	40102	'x[86,4,36]'	1	76290	'x[163,5,44]'	1
4527 'x[10,5,38]'	1	40771	'x[87,7,35]'	1	76686	'x[164,4,38]'	1
4836 'x[11,3,12]'	1	41018	'x[88,4,14]'	1	77352	'x[165,7,34]'	1
5410 'x[12,4,50]'	1	41548	'x[89,5,8]'	1	77744	'x[166,6,24]'	1
5641 'x[13,1,13]'	1	41985	'x[90,4,43]'	1	78263	'x[167,7,7]'	1
6502 'x[14,7,3]'	1	42335	'x[91,2,58]'	1	78479	'x[168,3,22]'	1
6964 'x[15,6,63]'	1	42962	'x[92,5,15]'	1	79189	'x[169,6,62]'	1
7300 'x[16,4,64]'	1	43428	'x[93,5,12]'	1	79317	'x[170,1,56]'	1
7609 'x[17,2,38]'	1	43887	'x[94,5,2]'	1	79922	'x[171,3,58]'	1
8081 'x[18,2,41]'	1	44421	'x[95,5,67]'	1	80371	'x[172,3,38]'	1
8861 'x[19,7,17]'	1	44898	'x[96,6,8]'	1	80992	'x[173,5,56]'	1
9131 'x[20,4,19]'	1	45412	'x[97,6,53]'	1	81416	'x[174,5,11]'	1
9567 'x[21,3,53]'	1	45938	'x[98,7,43]'	1	81772	'x[175,3,32]'	1
10262 'x[22,7,11]'	1	46151	'x[99,3,55]'	1	82404	'x[176,5,61]'	1
10332 'x[23,1,14]'	1	46688	'x[100,4,56]'	1	82630	'x[177,2,19]'	1
11025 'x[24,4,37]'	1	47130	'x[101,4,29]'	1	83352	'x[178,6,4]'	1
11542 'x[25,5,18]'	1	47384	'x[102,1,15]'	1	83634	'x[179,3,18]'	1

84344	'x[180,6,58]'	1	120351	'x[257,5,19]'	1	156198	'x[334,1,21]'	1
84889	'x[181,7,67]'	1	120841	'x[258,5,40]'	1	156740	'x[335,2,27]'	1
85233	'x[182,6,9]'	1	121230	'x[259,4,27]'	1	157360	'x[336,4,44]'	1
85814	'x[183,7,54]'	1	121536	'x[260,1,65]'	1	157916	'x[337,5,64]'	1
85953	'x[184,2,59]'	1	121964	'x[261,1,24]'	1	158065	'x[338,1,12]'	1
86630	'x[185,5,66]'	1	122570	'x[262,3,27]'	1	158739	'x[339,4,16]'	1
87218	'x[186,7,51]'	1	123338	'x[263,7,58]'	1	159449	'x[340,7,56]'	1
87688	'x[187,7,52]'	1	123487	'x[264,3,6]'	1	159639	'x[341,3,45]'	1
88034	'x[188,5,63]'	1	123821	'x[265,1,5]'	1	159995	'x[342,1,66]'	1
88535	'x[189,6,28]'	1	124389	'x[266,2,37]'	1	160816	'x[343,7,16]'	1
88733	'x[190,2,25]'	1	124934	'x[267,3,46]'	1	161197	'x[344,5,62]'	1
89226	'x[191,2,49]'	1	125511	'x[268,5,20]'	1	161533	'x[345,3,63]'	1
89690	'x[192,2,44]'	1	125849	'x[269,3,23]'	1	162083	'x[346,5,10]'	1
90459	'x[193,7,9]'	1	126270	'x[270,2,42]'	1	162469	'x[347,3,61]'	1
90927	'x[194,7,8]'	1	126729	'x[271,2,32]'	1	162839	'x[348,2,29]'	1
91118	'x[195,2,65]'	1	127401	'x[272,5,34]'	1	163678	'x[349,7,64]'	1
91598	'x[196,3,9]'	1	127822	'x[273,4,53]'	1	163740	'x[350,1,59]'	1
92217	'x[197,5,25]'	1	128501	'x[274,7,62]'	1	164441	'x[351,5,23]'	1
92552	'x[198,3,25]'	1	128715	'x[275,4,8]'	1	165004	'x[352,6,50]'	1
93296	'x[199,7,32]'	1	129321	'x[276,6,11]'	1	165414	'x[353,5,58]'	1
93757	'x[200,7,24]'	1	129696	'x[277,4,51]'	1	165813	'x[354,4,55]'	1
94120	'x[201,5,52]'	1	130140	'x[278,4,26]'	1	166139	'x[355,2,46]'	1
94565	'x[202,5,28]'	1	130822	'x[279,7,38]'	1	166528	'x[356,1,33]'	1
94873	'x[203,3,1]'	1	131306	'x[280,7,53]'	1	167274	'x[357,5,42]'	1
95496	'x[204,5,21]'	1	131567	'x[281,4,46]'	1	167744	'x[358,5,43]'	1
95790	'x[205,2,47]'	1	132138	'x[282,6,14]'	1	168227	'x[359,5,57]'	1
96416	'x[206,5,3]'	1	132660	'x[283,6,67]'	1	168729	'x[360,6,23]'	1
96651	'x[207,1,37]'	1	132885	'x[284,3,24]'	1	169291	'x[361,7,49]'	1
97466	'x[208,6,48]'	1	133510	'x[285,5,46]'	1	169404	'x[362,2,28]'	1
97759	'x[209,4,6]'	1	134006	'x[286,6,6]'	1	170157	'x[363,6,44]'	1
98103	'x[210,2,15]'	1	134435	'x[287,5,33]'	1	170587	'x[364,6,5]'	1
98951	'x[211,7,59]'	1	134678	'x[288,2,8]'	1	170756	'x[365,1,40]'	1
99014	'x[212,1,55]'	1	135364	'x[289,5,24]'	1	171575	'x[366,6,55]'	1
99832	'x[213,7,2]'	1	135745	'x[290,4,3]'	1	172025	'x[367,6,36]'	1
100038	'x[214,3,7]'	1	136331	'x[291,5,53]'	1	172127	'x[368,1,4]'	1
100766	'x[215,6,65]'	1	136849	'x[292,6,35]'	1	172970	'x[369,6,43]'	1
101130	'x[216,5,27]'	1	137130	'x[293,3,48]'	1	173450	'x[370,6,54]'	1
101705	'x[217,6,66]'	1	137469	'x[294,1,52]'	1	173633	'x[371,2,36]'	1
102155	'x[218,6,47]'	1	138315	'x[295,7,27]'	1	174190	'x[372,3,57]'	1
102384	'x[219,3,8]'	1	138579	'x[296,4,23]'	1	174773	'x[373,5,37]'	1
102804	'x[220,2,26]'	1	138989	'x[297,3,31]'	1	175380	'x[374,7,41]'	1
103601	'x[221,7,19]'	1	139507	'x[298,4,13]'	1	175489	'x[375,2,16]'	1
103771	'x[222,2,55]'	1	140078	'x[299,5,48]'	1	176281	'x[376,7,4]'	1
104255	'x[223,3,3]'	1	140521	'x[300,5,22]'	1	176532	'x[377,3,54]'	1
104939	'x[224,6,17]'	1	140806	'x[301,2,39]'	1	176934	'x[378,2,54]'	1
105118	'x[225,1,62]'	1	141266	'x[302,2,30]'	1	177492	'x[379,4,9]'	1
105887	'x[226,6,27]'	1	142003	'x[303,6,30]'	1	177949	'x[380,3,64]'	1
106371	'x[227,6,42]'	1	142499	'x[304,6,57]'	1	178556	'x[381,6,1]'	1
106502	'x[228,1,39]'	1	142792	'x[305,4,15]'	1	179138	'x[382,7,47]'	1
107191	'x[229,4,58]'	1	143420	'x[306,6,40]'	1	179305	'x[383,3,13]'	1
107758	'x[230,6,22]'	1	143555	'x[307,1,41]'	1	179805	'x[384,3,44]'	1
107886	'x[231,1,16]'	1	144208	'x[308,4,24]'	1	180260	'x[385,3,30]'	1
108694	'x[232,6,20]'	1	144623	'x[309,3,37]'	1	180925	'x[386,6,25]'	1
109026	'x[233,4,17]'	1	144964	'x[310,1,43]'	1	181361	'x[387,5,59]'	1
109280	'x[234,1,3]'	1	145500	'x[311,2,43]'	1	181632	'x[388,2,62]'	1
110008	'x[235,4,61]'	1	145897	'x[312,1,38]'	1	182424	'x[389,7,50]'	1
110490	'x[236,5,7]'	1	146758	'x[313,7,28]'	1	182634	'x[390,3,59]'	1
110817	'x[237,2,66]'	1	147192	'x[314,6,60]'	1	183290	'x[391,6,45]'	1
111329	'x[238,3,42]'	1	147539	'x[315,5,5]'	1	183541	'x[392,3,28]'	1
111986	'x[239,6,29]'	1	148183	'x[316,7,46]'	1	184061	'x[393,4,12]'	1
112162	'x[240,2,4]'	1	148410	'x[317,4,5]'	1	184353	'x[394,1,36]'	1
112977	'x[241,7,15]'	1	148921	'x[318,4,47]'	1	185180	'x[395,6,59]'	1
113367	'x[242,6,3]'	1	149601	'x[319,7,57]'	1	185641	'x[396,6,51]'	1
113882	'x[243,6,49]'	1	150038	'x[320,7,25]'	1	185958	'x[397,4,33]'	1
114087	'x[244,2,53]'	1	150526	'x[321,7,44]'	1	186251	'x[398,1,58]'	1
114659	'x[245,4,22]'	1	150864	'x[322,5,47]'	1	186807	'x[399,3,11]'	1
114928	'x[246,1,23]'	1	151220	'x[323,4,1]'	1	187364	'x[400,4,32]'	1
115458	'x[247,2,17]'	1	151489	'x[324,1,2]'	1	188065	'x[401,7,63]'	1
116209	'x[248,6,31]'	1	152159	'x[325,4,2]'	1	188504	'x[402,7,33]'	1
116575	'x[249,4,62]'	1	152647	'x[326,4,21]'	1	188946	'x[403,7,6]'	1
117222	'x[250,7,39]'	1	153126	'x[327,4,31]'	1	189388	'x[404,6,46]'	1
117553	'x[251,5,35]'	1	153630	'x[328,4,66]'	1	189518	'x[405,1,42]'	1
118147	'x[252,7,26]'	1	154257	'x[329,7,23]'	1	190043	'x[406,2,31]'	1
118217	'x[253,1,29]'	1	154500	'x[330,3,65]'	1	190675	'x[407,4,60]'	1
118934	'x[254,5,9]'	1	155016	'x[331,4,45]'	1	190913	'x[408,1,30]'	1
119362	'x[255,4,35]'	1	155635	'x[332,6,61]'	1	191573	'x[409,4,20]'	1
119986	'x[256,6,56]'	1	155862	'x[333,3,20]'	1	192119	'x[410,5,30]'	1

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192291 'x[411,1,1]' 1 202320 'x[432,3,47]' 1 212355 'x[453,6,32]' 1
193206 'x[412,7,45]' 1 202907 'x[433,5,31]' 1 212920 'x[454,7,61]' 1
193471 'x[413,4,42]' 1 203346 'x[434,5,1]' 1 213050 'x[455,2,57]' 1
193865 'x[414,3,34]' 1 203685 'x[435,3,5]' 1 213661 'x[456,4,65]' 1
194317 'x[415,3,17]' 1 204130 'x[436,2,48]' 1 214060 'x[457,3,62]' 1
194788 'x[416,3,19]' 1 204647 'x[437,3,29]' 1 214666 'x[458,5,65]' 1
195194 'x[417,2,23]' 1 205030 'x[438,2,10]' 1 214890 'x[459,2,21]' 1
195600 'x[418,1,27]' 1 205719 'x[439,5,29]' 1 215618 'x[460,6,12]' 1
196284 'x[419,4,41]' 1 205926 'x[440,1,35]' 1 215989 'x[461,4,48]' 1
196914 'x[420,7,1]' 1 206641 'x[441,5,13]' 1 216231 'x[462,1,22]' 1
196990 'x[421,1,10]' 1 207252 'x[442,7,21]' 1 216942 'x[463,4,63]' 1
197756 'x[422,5,39]' 1 207378 'x[443,2,13]' 1 217497 'x[464,6,15]' 1
198005 'x[423,2,20]' 1 207953 'x[444,3,52]' 1 217680 'x[465,1,64]' 1
198740 'x[424,6,18]' 1 208355 'x[445,2,52]' 1 218102 'x[466,1,17]' 1
198979 'x[425,2,56]' 1 209137 'x[446,7,30]' 1 218682 'x[467,2,61]' 1
199382 'x[426,1,57]' 1 209193 'x[447,1,19]' 1 219072 'x[468,1,49]' 1
200052 'x[427,4,57]' 1 210011 'x[448,6,33]' 1 219537 'x[469,1,45]' 1
200331 'x[428,2,1]' 1 210184 'x[449,2,5]' 1 ;
200739 'x[429,1,7]' 1 210627 'x[450,1,46]' 1
201286 'x[430,2,18]' 1 211235 'x[451,3,51]' 1 :_varname[j] _var[j] :=
201679 'x[431,1,9]' 1 211527 'x[452,1,8]' 1 ;

```

Fase II – (Variante) Problema de Afetação Generalizado (Binário)

Nesta fase já é permitido armazenar mais que uma paleta por alvéolo. Tendo em conta que cada alvéolo tem uma largura de 3,2 metros, ou seja, a largura total das paletes usadas não pode ser superior a 3,2 metros, sendo esta a restrição para esta fase.

A tabela abaixo representa a disposição possível de paletes nos alvéolos, sendo X (1) o número de paletes do tipo industrial, X (2) o número de paletes do tipo europeu e Total o número de metros ocupados pelo respetivo número de paletes do tipo industrial e europeu.

X (1)	0	3	2	1	0
X (2)	4	0	1	2	3
Total	3.2	3	2.8	2.6	2.4

Como nos é pedido no enunciado para minimizar a área disponível para armazenamento de paletes no armazém, e nos foi dada a liberdade de escolher o layout do armazém, optamos por um layout que fosse de encontro a minimizar a área disponível.

Como apenas há 47 paletes europeias, e sendo que estas ocupam a área total de cada alvéolo quando agrupadas em 4 ($4 \times 0.8 = 3.2$), agrupamos 44 paletes por 11 alvéolos, mas optamos por que cada *rack* tivesse 10 alvéolos porque questões de estética. Ficando assim 4 paletes europeias postas de parte, visto que a 2ª disposição que desperdiça menos área é a de 3 paletes industriais, sendo assim mais benéfico agrupar posteriormente as 4 paletes europeias juntas.

Assim, temos agora de agrupar as 422 paletes industriais.

Se dividirmos as paletes por *racks* de 10 alvéolos, cada *rack* albergaria 30 paletes, sendo $422/30 = 14$ (quociente inteiro). Dispõe-se assim de 14 *racks* de 10 níveis para albergar 420 paletes ($14 \times 30 = 420$), sobrando 2 paletes industriais.

Sobram assim 7 paletes europeias e 2 industriais. 4 paletes europeias têm de ficar sozinhas, pois a área desperdiçada é nula, sobrando assim 3 paletes europeias e 2 industriais. Sendo neste momento irrelevante como dispomos estas últimas 5, visto que são necessários 2 alvéolos qualquer que seja a disposição. Assim para estas últimas paletes e como nos foi dada a liberdade de escolher *racks* com diferente número de alvéolos.

Assim, o número total de *racks* usado é de: 15 *racks* de 10 alvéolos e 1 *rack* de 3 alvéolos.

Com isto a área disponível é mínima, albergando todas as 469 paletes.

```
***** 469 inequality 15530 'x[34,1,53]' 1
```

1 - 8

Corredor 1

9 - 16

```
***** constraints 15983 'x[35,1,37]' 1
***** 1 linear objective; 219961 16468 'x[36,1,53]' 1
nonzeros. 16921 'x[37,1,37]' 1
17371 'x[38,1,18]' 1
NEOS Server Version 5.0 CPLEX 12.7.0.0: threads=4 17858 'x[39,1,36]' 1
Job# : 6106702 CPLEX 12.7.0.0: optimal 18350 'x[40,1,59]' 1
Password : LMubAODG integer solution within 18779 'x[41,1,19]' 1
User : None mipgap or absmipgap; 19248 'x[42,1,19]' 1
Solver : objective 24827 19718 'x[43,1,20]' 1
milp:CPLEX:AMPL 302939 MIP simplex 20356 'x[44,3,55]' 1
Start : 2018-06-06 iterations 20670 'x[45,1,34]' 1
12:31:29 11938 branch-and-bound nodes 21164 'x[46,1,59]' 1
End : 2018-06-06 absmipgap = 2.25, relmipgap 21608 'x[47,1,34]' 1
12:37:20 = 9.06271e-05 22070 'x[48,1,27]' 1
Host : NEOS HTCondor No basis. 22619 'x[49,2,40]' 1
Pool : _varname[j] 23069 'x[50,2,21]' 1
Disclaimer: _var[j] := 23481 'x[51,1,31]' 1
158 'x[1,3,24]' 1 24056 'x[52,3,3]' 1
This information is 487 'x[2,1,18]' 1 24449 'x[53,1,61]' 1
provided without any express 954 'x[3,1,16]' 1 25229 'x[54,6,37]' 1
or 1423 'x[4,1,16]' 1 25680 'x[55,6,19]' 1
implied warranty. In 1896 'x[5,1,20]' 1 25949 'x[56,3,20]' 1
particular, there is no 2416 'x[6,2,4]' 1 26327 'x[57,1,63]' 1
warranty 2972 'x[7,3,24]' 1 26755 'x[58,1,22]' 1
of any kind concerning 3303 'x[8,1,20]' 1 27507 'x[59,5,37]' 1
the fitness of this 3768 'x[9,1,16]' 1 27716 'x[60,1,45]' 1
information for any 4274 'x[10,1,53]' 1 28444 'x[61,5,36]' 1
particular purpose. 4749 'x[11,1,59]' 1 28614 'x[62,1,5]' 1
***** 5230 'x[12,2,4]' 1 29183 'x[63,2,38]' 1
***** 5665 'x[13,1,37]' 1 29575 'x[64,1,28]' 1
***** 6156 'x[14,1,59]' 1 30173 'x[65,3,23]' 1
File exists 7005 'x[15,7,37]' 1 30542 'x[66,1,57]' 1
You are using the solver 7069 'x[16,1,34]' 1 31026 'x[67,2,5]' 1
cplexamp. 7540 'x[17,1,36]' 1 31843 'x[68,7,18]' 1
Checking ampl.mod for 8010 'x[18,1,37]' 1 31948 'x[69,1,56]' 1
cplex_options... 8600 'x[19,3,24]' 1 32425 'x[70,1,64]' 1
Checking ampl.com for 8929 'x[20,1,18]' 1 32887 'x[71,1,57]' 1
cplex_options... 9400 'x[21,1,20]' 1 33343 'x[72,1,44]' 1
Executing AMPL. 10007 'x[22,3,24]' 1 33842 'x[73,2,7]' 1
processing data. 10371 'x[23,1,53]' 1 34394 'x[74,3,23]' 1
processing commands. 10806 'x[24,1,19]' 1 34793 'x[75,2,20]' 1
Executing on prod-exec- 11275 'x[25,1,19]' 1 35350 'x[76,3,41]' 1
5.neos-server.org 11743 'x[26,1,18]' 1 35693 'x[77,1,49]' 1
12230 'x[27,1,36]' 1 36332 'x[78,4,18]' 1
219961 variables, all binary 12734 'x[28,2,4]' 1 36623 'x[79,1,41]' 1
938 constraints, all linear; 13571 'x[29,7,37]' 1 37422 'x[80,6,36]' 1
439922 nonzeros 13635 'x[30,1,34]' 1 37531 'x[81,1,11]' 1
469 equality 14086 'x[31,1,16]' 1 38109 'x[82,2,53]' 1
constraints 14610 'x[32,2,4]' 1 38523 'x[83,1,65]' 1
15044 'x[33,1,36]' 1 39019 'x[84,2,25]' 1
```

39834	'x[85,7,36]'	1	75632	'x[162,2,56]'	1	111876	'x[239,4,53]'	1
39968	'x[86,2,36]'	1	75989	'x[163,1,11]'	1	112262	'x[240,3,37]'	1
40471	'x[87,3,3]'	1	76618	'x[164,3,37]'	1	112634	'x[241,2,7]'	1
41025	'x[88,4,21]'	1	77042	'x[165,2,59]'	1	113042	'x[242,1,13]'	1
41364	'x[89,2,25]'	1	77455	'x[166,2,3]'	1	113610	'x[243,2,45]'	1
41803	'x[90,1,62]'	1	78275	'x[167,7,19]'	1	114070	'x[244,2,36]'	1
42235	'x[91,1,25]'	1	78411	'x[168,2,21]'	1	114443	'x[245,1,7]'	1
43118	'x[92,7,37]'	1	78865	'x[169,2,6]'	1	114945	'x[246,1,40]'	1
43569	'x[93,7,19]'	1	79283	'x[170,1,22]'	1	115383	'x[247,1,9]'	1
43737	'x[94,2,53]'	1	79776	'x[171,1,46]'	1	116033	'x[248,3,56]'	1
44161	'x[95,2,8]'	1	80254	'x[172,1,55]'	1	116384	'x[249,2,5]'	1
44628	'x[96,2,6]'	1	80823	'x[173,3,21]'	1	116871	'x[250,2,23]'	1
45395	'x[97,6,36]'	1	81262	'x[174,2,58]'	1	117538	'x[251,5,20]'	1
45559	'x[98,1,66]'	1	81845	'x[175,4,38]'	1	117786	'x[252,1,67]'	1
45967	'x[99,1,5]'	1	82126	'x[176,1,51]'	1	118230	'x[253,1,42]'	1
46462	'x[100,1,31]'	1	82714	'x[177,3,36]'	1	118831	'x[254,3,40]'	1
47035	'x[101,3,1]'	1	83300	'x[178,5,19]'	1	119300	'x[255,3,40]'	1
47588	'x[102,4,18]'	1	83484	'x[179,1,2]'	1	119627	'x[256,1,32]'	1
47963	'x[103,2,58]'	1	83968	'x[180,1,17]'	1	120088	'x[257,1,24]'	1
48318	'x[104,1,11]'	1	84546	'x[181,2,59]'	1	120541	'x[258,1,8]'	1
48851	'x[105,2,8]'	1	85091	'x[182,4,1]'	1	121111	'x[259,2,42]'	1
49311	'x[106,1,66]'	1	85405	'x[183,1,47]'	1	121529	'x[260,1,58]'	1
49901	'x[107,3,53]'	1	85947	'x[184,2,53]'	1	122092	'x[261,3,18]'	1
50277	'x[108,2,27]'	1	86348	'x[185,1,52]'	1	122469	'x[262,1,60]'	1
50720	'x[109,2,1]'	1	86968	'x[186,4,2]'	1	122892	'x[263,1,14]'	1
51134	'x[110,1,13]'	1	87475	'x[187,4,40]'	1	123353	'x[264,1,6]'	1
51644	'x[111,1,54]'	1	87755	'x[188,1,52]'	1	123902	'x[265,2,19]'	1
52105	'x[112,1,46]'	1	88412	'x[189,4,39]'	1	124325	'x[266,1,40]'	1
52814	'x[113,5,18]'	1	88698	'x[190,1,57]'	1	124802	'x[267,1,48]'	1
53153	'x[114,3,22]'	1	89330	'x[191,4,19]'	1	125394	'x[268,3,37]'	1
53499	'x[115,1,33]'	1	89637	'x[192,1,58]'	1	125996	'x[269,5,36]'	1
53985	'x[116,1,50]'	1	90271	'x[193,4,22]'	1	126217	'x[270,1,56]'	1
54430	'x[117,1,26]'	1	90771	'x[194,4,53]'	1	126693	'x[271,1,63]'	1
55062	'x[118,3,55]'	1	91048	'x[195,1,62]'	1	127210	'x[272,2,44]'	1
55349	'x[119,1,7]'	1	91485	'x[196,1,30]'	1	127593	'x[273,1,25]'	1
56184	'x[120,6,38]'	1	92059	'x[197,3,1]'	1	128260	'x[274,4,22]'	1
56332	'x[121,1,52]'	1	92764	'x[198,6,36]'	1	128518	'x[275,1,12]'	1
56856	'x[122,2,40]'	1	92975	'x[199,2,46]'	1	129102	'x[276,2,60]'	1
57256	'x[123,1,38]'	1	93440	'x[200,2,42]'	1	129684	'x[277,4,39]'	1
57697	'x[124,1,10]'	1	93976	'x[201,3,42]'	1	130004	'x[278,2,24]'	1
58197	'x[125,1,41]'	1	94623	'x[202,6,19]'	1	130735	'x[279,6,18]'	1
58716	'x[126,2,24]'	1	94793	'x[203,1,55]'	1	131038	'x[280,3,53]'	1
59221	'x[127,2,60]'	1	95445	'x[204,4,37]'	1	131507	'x[281,3,53]'	1
59567	'x[128,1,4]'	1	95745	'x[205,2,2]'	1	131902	'x[282,2,46]'	1
60075	'x[129,1,43]'	1	96192	'x[206,1,47]'	1	132352	'x[283,2,27]'	1
60505	'x[130,1,4]'	1	96653	'x[207,1,39]'	1	132833	'x[284,2,39]'	1
61207	'x[131,4,36]'	1	97389	'x[208,5,38]'	1	133213	'x[285,1,17]'	1
61595	'x[132,3,22]'	1	97599	'x[209,1,47]'	1	133679	'x[286,1,14]'	1
61916	'x[133,1,8]'	1	98030	'x[210,1,9]'	1	134374	'x[287,4,39]'	1
62490	'x[134,2,46]'	1	98557	'x[211,1,67]'	1	134648	'x[288,1,45]'	1
63065	'x[135,4,18]'	1	99045	'x[212,2,19]'	1	135180	'x[289,2,41]'	1
63353	'x[136,1,38]'	1	99518	'x[213,2,23]'	1	135586	'x[290,1,45]'	1
63869	'x[137,2,18]'	1	100052	'x[214,3,21]'	1	136079	'x[291,2,2]'	1
64301	'x[138,1,48]'	1	100504	'x[215,3,4]'	1	136668	'x[292,3,55]'	1
64830	'x[139,2,41]'	1	101023	'x[216,3,54]'	1	136960	'x[293,1,12]'	1
65245	'x[140,1,54]'	1	101494	'x[217,3,56]'	1	137523	'x[294,2,39]'	1
65814	'x[141,3,20]'	1	102061	'x[218,5,20]'	1	138088	'x[295,4,1]'	1
66237	'x[142,2,41]'	1	102364	'x[219,2,55]'	1	138476	'x[296,2,54]'	1
66649	'x[143,1,51]'	1	102864	'x[220,3,19]'	1	138997	'x[297,3,39]'	1
67129	'x[144,1,62]'	1	103286	'x[221,2,39]'	1	139429	'x[298,3,2]'	1
67629	'x[145,2,26]'	1	103691	'x[222,1,42]'	1	139886	'x[299,2,57]'	1
68160	'x[146,3,21]'	1	104186	'x[223,2,1]'	1	140238	'x[300,1,7]'	1
68846	'x[147,6,37]'	1	104763	'x[224,3,42]'	1	140756	'x[301,1,56]'	1
68984	'x[148,1,41]'	1	105293	'x[225,4,36]'	1	141229	'x[302,1,60]'	1
69584	'x[149,3,38]'	1	105681	'x[226,3,22]'	1	141653	'x[303,1,15]'	1
69884	'x[150,1,3]'	1	106299	'x[227,5,37]'	1	142142	'x[304,1,35]'	1
70393	'x[151,1,43]'	1	106566	'x[228,2,36]'	1	142643	'x[305,1,67]'	1
70852	'x[152,1,33]'	1	107370	'x[229,7,36]'	1	143054	'x[306,1,9]'	1
71290	'x[153,1,2]'	1	107523	'x[230,2,55]'	1	143599	'x[307,2,18]'	1
71929	'x[154,3,38]'	1	107896	'x[231,1,26]'	1	144052	'x[308,2,2]'	1
72579	'x[155,6,18]'	1	108399	'x[232,1,60]'	1	144473	'x[309,1,21]'	1
72745	'x[156,1,50]'	1	109229	'x[233,7,19]'	1	145007	'x[310,2,19]'	1
73253	'x[157,2,22]'	1	109278	'x[234,1,1]'	1	145416	'x[311,1,26]'	1
73744	'x[158,2,44]'	1	109795	'x[235,1,49]'	1	145899	'x[312,1,40]'	1
74305	'x[159,4,2]'	1	110569	'x[236,6,19]'	1	146401	'x[313,2,6]'	1
74683	'x[160,2,45]'	1	110774	'x[237,2,23]'	1	146890	'x[314,2,26]'	1
75261	'x[161,4,20]'	1	111373	'x[238,4,19]'	1	147487	'x[315,4,20]'	1

147846	'x[316,2,44]'	1	172641	'x[369,1,49]'	1	197477	'x[422,1,28]'	1
148624	'x[317,7,18]'	1	173131	'x[370,2,3]'	1	198003	'x[423,2,18]'	1
148960	'x[318,5,19]'	1	173551	'x[371,1,21]'	1	198557	'x[424,3,36]'	1
149266	'x[319,2,57]'	1	174026	'x[372,1,27]'	1	198899	'x[425,1,43]'	1
149735	'x[320,2,57]'	1	174620	'x[373,3,18]'	1	199326	'x[426,1,1]'	1
150095	'x[321,1,15]'	1	175110	'x[374,3,39]'	1	199804	'x[427,1,10]'	1
150563	'x[322,1,14]'	1	175454	'x[375,1,48]'	1	200317	'x[428,1,54]'	1
151023	'x[323,1,5]'	1	176063	'x[376,3,54]'	1	200756	'x[429,1,24]'	1
151591	'x[324,2,37]'	1	176388	'x[377,1,44]'	1	201223	'x[430,1,22]'	1
151981	'x[325,1,25]'	1	176868	'x[378,1,55]'	1	201775	'x[431,2,38]'	1
152729	'x[326,5,36]'	1	177521	'x[379,4,38]'	1	202559	'x[432,7,18]'	1
153004	'x[327,2,43]'	1	177886	'x[380,3,1]'	1	202914	'x[433,5,38]'	1
153488	'x[328,2,58]'	1	178228	'x[381,1,8]'	1	203184	'x[434,2,40]'	1
153955	'x[329,2,56]'	1	178763	'x[382,2,7]'	1	203667	'x[435,2,54]'	1
154332	'x[330,1,31]'	1	179221	'x[383,1,63]'	1	204169	'x[436,3,20]'	1
155057	'x[331,5,19]'	1	179716	'x[384,2,22]'	1	204513	'x[437,1,29]'	1
155461	'x[332,4,21]'	1	180218	'x[385,2,55]'	1	204980	'x[438,1,27]'	1
155795	'x[333,2,20]'	1	180688	'x[386,2,56]'	1	205514	'x[439,2,25]'	1
156215	'x[334,1,38]'	1	181051	'x[387,1,17]'	1	206177	'x[440,5,18]'	1
156704	'x[335,1,58]'	1	181741	'x[388,4,37]'	1	206648	'x[441,5,20]'	1
157144	'x[336,1,29]'	1	182148	'x[389,3,42]'	1	206859	'x[442,1,30]'	1
157806	'x[337,4,21]'	1	182471	'x[390,1,30]'	1	207344	'x[443,1,46]'	1
158206	'x[338,3,19]'	1	183282	'x[391,6,37]'	1	207877	'x[444,2,43]'	1
158828	'x[339,5,38]'	1	183817	'x[392,7,36]'	1	208265	'x[445,1,29]'	1
159127	'x[340,3,2]'	1	183913	'x[393,1,65]'	1	208800	'x[446,2,28]'	1
159473	'x[341,1,13]'	1	184356	'x[394,1,39]'	1	209175	'x[447,1,1]'	1
159935	'x[342,1,6]'	1	184852	'x[395,1,66]'	1	209884	'x[448,4,40]'	1
160473	'x[343,2,8]'	1	185430	'x[396,3,41]'	1	210118	'x[449,1,6]'	1
161069	'x[344,4,1]'	1	185775	'x[397,1,51]'	1	210818	'x[450,4,36]'	1
161400	'x[345,1,64]'	1	186297	'x[398,2,37]'	1	211355	'x[451,5,37]'	1
161941	'x[346,3,2]'	1	186726	'x[399,1,64]'	1	211624	'x[452,2,38]'	1
162335	'x[347,1,61]'	1	187319	'x[400,3,54]'	1	212243	'x[453,4,54]'	1
162963	'x[348,4,19]'	1	187775	'x[401,3,41]'	1	212551	'x[454,2,27]'	1
163307	'x[349,2,28]'	1	188442	'x[402,6,38]'	1	213079	'x[455,3,19]'	1
163704	'x[350,1,23]'	1	188712	'x[403,3,40]'	1	213467	'x[456,2,5]'	1
164241	'x[351,2,24]'	1	189261	'x[404,4,53]'	1	213896	'x[457,1,32]'	1
164874	'x[352,4,54]'	1	189646	'x[405,3,36]'	1	214348	'x[458,1,15]'	1
165121	'x[353,1,33]'	1	190034	'x[406,2,22]'	1	214826	'x[459,1,24]'	1
165585	'x[354,1,28]'	1	190449	'x[407,1,35]'	1	215397	'x[460,2,59]'	1
166264	'x[355,4,37]'	1	190886	'x[408,1,3]'	1	215805	'x[461,1,65]'	1
166499	'x[356,1,4]'	1	191420	'x[409,2,1]'	1	216361	'x[462,3,18]'	1
167025	'x[357,1,61]'	1	191958	'x[410,3,3]'	1	216788	'x[463,2,43]'	1
167445	'x[358,1,12]'	1	192394	'x[411,2,37]'	1	217259	'x[464,2,45]'	1
167934	'x[359,1,32]'	1	192852	'x[412,2,26]'	1	217704	'x[465,2,21]'	1
168610	'x[360,4,38]'	1	193238	'x[413,1,10]'	1	218129	'x[466,1,44]'	1
168997	'x[361,3,23]'	1	193747	'x[414,1,50]'	1	218907	'x[467,6,18]'	1
169430	'x[362,2,54]'	1	194201	'x[415,1,35]'	1	219309	'x[468,5,18]'	1
169887	'x[363,2,42]'	1	194656	'x[416,1,21]'	1	219515	'x[469,1,23]'	1
170468	'x[364,4,20]'	1	195146	'x[417,1,42]'	1	;		
170718	'x[365,1,2]'	1	195576	'x[418,1,3]'	1			
171358	'x[366,3,39]'	1	196214	'x[419,3,38]'	1	:_varname[j] _var[j]	:=	
171741	'x[367,2,20]'	1	196581	'x[420,2,3]'	1	;		
172162	'x[368,1,39]'	1	197003	'x[421,1,23]'	1			

Figura 3 - Disposição do armazém na fase II.

Fase III – Problema de Afetação e Layout de Armazém

Nesta fase já é tido em conta o parâmetro z (altura) das paletes e A (altura) dos alvéolos. Sendo que não é possível armazenar paletes umas em cima das outras, e que uma paleta

tem no máximo 1 metro (parâmetro altura da figura 1, que apesar de não conter unidades, foi nos dito pelo docente que a medida estava em centímetros). E como no nosso caso os alvéolos têm altura de 1 metro (parâmetro escolhido arbitrariamente, como nos permite o enunciado).

Visto isto, não conseguimos perceber o intuito desta fase, pois o trabalho de otimizar a área disponível (minimizando), já tinha sido feito na segunda fase, e não seria necessário otimizar nada, sendo que a “área vertical” (altura*largura, como está explícito a certa altura no enunciado), teria um valor constante.

E as paletes com o parâmetro W (utilização anual) mais elevado devem estar presentes numa *rack* mais próxima da porta, pois a distância percorrida pelos operadores vai ser menor, otimizando distâncias e tempos.

Data:

param	NPALETES:=469;	21	1	1	0.8	7
param	NPRATELEIRAS:=7;	22	1	1	0.8	1
param	NESTANTES:=67;	23	1	1	0.8	7
		24	1	1	0.8	8
param	PALETE:	25	1	1	0.8	8
	1 2 3 4:=	26	1	1	0.8	8
1	1 1 0.8 1	27	1	1	0.8	8
2	1 1 0.8 9	28	1	1	0.8	3
3	1 1 0.8 1	29	1	1	0.8	1
4	1 1 0.8 2	30	1	1	0.8	4
5	1 1 0.8 7	31	1	1	0.8	2
6	1 1 0.8 3	32	1	1	0.8	3
7	1 1 0.8 1	33	1	1	0.8	8
8	1 1 0.8 7	34	1	1	0.8	7
9	1 1 0.8 2	35	1	1	0.8	8
10	1 1 0.8 7	36	1	1	0.8	6
11	1 1 0.8 4	37	1	1	0.8	7
12	1 1 0.8 3	38	1	1	0.8	8
13	1 1 0.8 8	39	1	1	0.8	9
14	1 1 0.8 4	40	1	1	0.8	5
15	1 1 0.8 1	41	1	1	0.8	8
16	1 1 0.8 3	42	1	1	0.8	8
17	1 1 0.8 9	43	1	1	0.8	7
18	1 1 0.8 8	44	1	1	0.8	1
19	1 1 0.8 1	45	1	1	0.8	4
20	1 1 0.8 8	46	1	1	0.8	5

47	1	1	0.8	3	102	1	1	1	7
48	1	1	1	6	103	1	1	1	2
49	1	1	1	6	104	1	1	1	4
50	1	1	1	7	105	1	1	1	1
51	1	1	1	5	106	1	1	1	3
52	1	1	1	1	107	1	1	1	4
53	1	1	1	5	108	1	1	1	1
54	1	1	1	3	109	1	1	1	7
55	1	1	1	3	110	1	1	1	5
56	1	1	1	5	111	1	1	1	8
57	1	1	1	4	112	1	1	1	6
58	1	1	1	9	113	1	1	1	7
59	1	1	1	5	114	1	1	1	3
60	1	1	1	6	115	1	1	1	4
61	1	1	1	7	116	1	1	1	4
62	1	1	1	7	117	1	1	1	7
63	1	1	1	8	118	1	1	1	1
64	1	1	1	6	119	1	1	1	6
65	1	1	1	1	120	1	1	1	1
66	1	1	1	7	121	1	1	1	2
67	1	1	1	3	122	1	1	1	6
68	1	1	1	5	123	1	1	1	9
69	1	1	1	8	124	1	1	1	5
70	1	1	1	4	125	1	1	1	8
71	1	1	1	7	126	1	1	1	5
72	1	1	1	7	127	1	1	1	1
73	1	1	1	1	128	1	1	1	8
74	1	1	1	2	129	1	1	1	7
75	1	1	1	8	130	1	1	1	8
76	1	1	1	1	131	1	1	1	8
77	1	1	1	4	132	1	1	1	3
78	1	1	1	7	133	1	1	1	6
79	1	1	1	8	134	1	1	1	1
80	1	1	1	6	135	1	1	1	8
81	1	1	1	4	136	1	1	1	9
82	1	1	1	7	137	1	1	1	9
83	1	1	1	3	138	1	1	1	5
84	1	1	1	3	139	1	1	1	5
85	1	1	1	5	140	1	1	1	9
86	1	1	1	9	141	1	1	1	6
87	1	1	1	1	142	1	1	1	5
88	1	1	1	2	143	1	1	1	3
89	1	1	1	3	144	1	1	1	5
90	1	1	1	5	145	1	1	1	2
91	1	1	1	7	146	1	1	1	5
92	1	1	1	1	147	1	1	1	3
93	1	1	1	2	148	1	1	1	8
94	1	1	1	6	149	1	1	1	6
95	1	1	1	1	150	1	1	1	8
96	1	1	1	2	151	1	1	1	7
97	1	1	1	6	152	1	1	1	4
98	1	1	1	3	153	1	1	1	9
99	1	1	1	7	154	1	1	1	6
100	1	1	1	5	155	1	1	1	6
101	1	1	1	4	156	1	1	1	4

157	1	1	1	6	212	1	1	1	9
158	1	1	1	2	213	1	1	1	5
159	1	1	1	1	214	1	1	1	5
160	1	1	1	1	215	1	1	1	1
161	1	1	1	4	216	1	1	1	3
162	1	1	1	4	217	1	1	1	1
163	1	1	1	4	218	1	1	1	2
164	1	1	1	7	219	1	1	1	5
165	1	1	1	1	220	1	1	1	7
166	1	1	1	5	221	1	1	1	7
167	1	1	1	2	222	1	1	1	8
168	1	1	1	7	223	1	1	1	7
169	1	1	1	2	224	1	1	1	1
170	1	1	1	9	225	1	1	1	8
171	1	1	1	6	226	1	1	1	3
172	1	1	1	8	227	1	1	1	4
173	1	1	1	5	228	1	1	1	9
174	1	1	1	2	229	1	1	1	5
175	1	1	1	4	230	1	1	1	5
176	1	1	1	3	231	1	1	1	7
177	1	1	1	9	232	1	1	1	6
178	1	1	1	4	233	1	1	1	2
179	1	1	1	9	234	1	1	1	9
180	1	1	1	3	235	1	1	1	4
181	1	1	1	1	236	1	1	1	3
182	1	1	1	2	237	1	1	1	5
183	1	1	1	5	238	1	1	1	6
184	1	1	1	7	239	1	1	1	2
185	1	1	1	2	240	1	1	1	7
186	1	1	1	1	241	1	1	1	1
187	1	1	1	1	242	1	1	1	5
188	1	1	1	2	243	1	1	1	1
189	1	1	1	2	244	1	1	1	9
190	1	1	1	7	245	1	1	1	6
191	1	1	1	6	246	1	1	1	9
192	1	1	1	7	247	1	1	1	5
193	1	1	1	1	248	1	1	1	1
194	1	1	1	2	249	1	1	1	3
195	1	1	1	5	250	1	1	1	5
196	1	1	1	5	251	1	1	1	2
197	1	1	1	4	252	1	1	1	2
198	1	1	1	6	253	1	1	1	8
199	1	1	1	1	254	1	1	1	3
200	1	1	1	4	255	1	1	1	3
201	1	1	1	1	256	1	1	1	4
202	1	1	1	3	257	1	1	1	8
203	1	1	1	8	258	1	1	1	6
204	1	1	1	6	259	1	1	1	4
205	1	1	1	6	260	1	1	1	7
206	1	1	1	5	261	1	1	1	9
207	1	1	1	9	262	1	1	1	5
208	1	1	1	2	263	1	1	1	3
209	1	1	1	5	264	1	1	1	6
210	1	1	1	5	265	1	1	1	9
211	1	1	1	2	266	1	1	1	9

267	1	1	1	5	322	1	1	1	3
268	1	1	1	7	323	1	1	1	7
269	1	1	1	7	324	1	1	1	9
270	1	1	1	7	325	1	1	1	7
271	1	1	1	5	326	1	1	1	7
272	1	1	1	2	327	1	1	1	3
273	1	1	1	7	328	1	1	1	2
274	1	1	1	1	329	1	1	1	4
275	1	1	1	4	330	1	1	1	4
276	1	1	1	1	331	1	1	1	4
277	1	1	1	2	332	1	1	1	2
278	1	1	1	5	333	1	1	1	8
279	1	1	1	6	334	1	1	1	9
280	1	1	1	5	335	1	1	1	7
281	1	1	1	4	336	1	1	1	5
282	1	1	1	1	337	1	1	1	2
283	1	1	1	1	338	1	1	1	7
284	1	1	1	7	339	1	1	1	2
285	1	1	1	3	340	1	1	1	3
286	1	1	1	3	341	1	1	1	5
287	1	1	1	2	342	1	1	1	7
288	1	1	1	6	343	1	1	1	1
289	1	1	1	5	344	1	1	1	2
290	1	1	1	6	345	1	1	1	4
291	1	1	1	6	346	1	1	1	3
292	1	1	1	1	347	1	1	1	5
293	1	1	1	4	348	1	1	1	6
294	1	1	1	7	349	1	1	1	1
295	1	1	1	2	350	1	1	1	8
296	1	1	1	6	351	1	1	1	5
297	1	1	1	4	352	1	1	1	1
298	1	1	1	3	353	1	1	1	4
299	1	1	1	3	354	1	1	1	6
300	1	1	1	6	355	1	1	1	6
301	1	1	1	8	356	1	1	1	7
302	1	1	1	6	357	1	1	1	5
303	1	1	1	2	358	1	1	1	4
304	1	1	1	4	359	1	1	1	4
305	1	1	1	2	360	1	1	1	4
306	1	1	1	5	361	1	1	1	1
307	1	1	1	9	362	1	1	1	6
308	1	1	1	6	363	1	1	1	3
309	1	1	1	9	364	1	1	1	4
310	1	1	1	9	365	1	1	1	9
311	1	1	1	7	366	1	1	1	4
312	1	1	1	9	367	1	1	1	8
313	1	1	1	2	368	1	1	1	9
314	1	1	1	2	369	1	1	1	4
315	1	1	1	4	370	1	1	1	5
316	1	1	1	2	371	1	1	1	9
317	1	1	1	5	372	1	1	1	6
318	1	1	1	4	373	1	1	1	8
319	1	1	1	3	374	1	1	1	4
320	1	1	1	3	375	1	1	1	5
321	1	1	1	2	376	1	1	1	3

377	1	1	1	7	432	1	1	1	5
378	1	1	1	8	433	1	1	1	2
379	1	1	1	4	434	1	1	1	6
380	1	1	1	4	435	1	1	1	6
381	1	1	1	6	436	1	1	1	6
382	1	1	1	1	437	1	1	1	5
383	1	1	1	5	438	1	1	1	6
384	1	1	1	6	439	1	1	1	3
385	1	1	1	5	440	1	1	1	7
386	1	1	1	4	441	1	1	1	2
387	1	1	1	3	442	1	1	1	5
388	1	1	1	6	443	1	1	1	6
389	1	1	1	1	444	1	1	1	3
390	1	1	1	5	445	1	1	1	5
391	1	1	1	3	446	1	1	1	1
392	1	1	1	5	447	1	1	1	9
393	1	1	1	3	448	1	1	1	1
394	1	1	1	9	449	1	1	1	7
395	1	1	1	3	450	1	1	1	8
396	1	1	1	1	451	1	1	1	4
397	1	1	1	3	452	1	1	1	8
398	1	1	1	9	453	1	1	1	1
399	1	1	1	4	454	1	1	1	1
400	1	1	1	3	455	1	1	1	7
401	1	1	1	1	456	1	1	1	3
402	1	1	1	1	457	1	1	1	4
403	1	1	1	3	458	1	1	1	2
404	1	1	1	2	459	1	1	1	8
405	1	1	1	9	460	1	1	1	1
406	1	1	1	6	461	1	1	1	3
407	1	1	1	4	462	1	1	1	9
408	1	1	1	8	463	1	1	1	3
409	1	1	1	7	464	1	1	1	1
410	1	1	1	2	465	1	1	1	7
411	1	1	1	9	466	1	1	1	7
412	1	1	1	2	467	1	1	1	6
413	1	1	1	5	468	1	1	1	7
414	1	1	1	4	469	1	1	1	8;
415	1	1	1	4					
416	1	1	1	9	param PRATELEIRAS:				
417	1	1	1	8	1 2 3 4:=				
418	1	1	1	8	1 1 1 1				
419	1	1	1	6	2 1 1 1				
420	1	1	1	5	3 1 1 1				
421	1	1	1	8	4 1 1 1				
422	1	1	1	6	5 1 1 1				
423	1	1	1	9	6 1 1 1				
424	1	1	1	8	7 1 1 1				
425	1	1	1	7					
426	1	1	1	9	param ESTANTES:				
427	1	1	1	5	1 2 3 4:=				
428	1	1	1	9	1 1 1 1				
429	1	1	1	8	2 1 1 1				
430	1	1	1	9	3 1 1 1				
431	1	1	1	8	4 1 1 1				

5	1	1	1	9	37	1	1	1	3
6	1	1	1	10	38	1	1	1	4
7	1	1	1	11	39	1	1	1	5
8	1	1	1	12	40	1	1	1	6
9	1	1	1	13	41	1	1	1	7
10	1	1	1	14	42	1	1	1	8
11	1	1	1	15	43	1	1	1	9
12	1	1	1	16	44	1	1	1	10
13	1	1	1	14	45	1	1	1	11
14	1	1	1	18	46	1	1	1	12
15	1	1	1	19	47	1	1	1	13
16	1	1	1	19	48	1	1	1	14
17	1	1	1	18	49	1	1	1	15
18	1	1	1	2	50	1	1	1	16
19	1	1	1	3	51	1	1	1	18
20	1	1	1	4	52	1	1	1	19
21	1	1	1	5	53	1	1	1	5
22	1	1	1	6	54	1	1	1	6
23	1	1	1	7	55	1	1	1	7
24	1	1	1	7	56	1	1	1	8
25	1	1	1	9	57	1	1	1	9
26	1	1	1	10	58	1	1	1	10
27	1	1	1	11	59	1	1	1	11
28	1	1	1	12	60	1	1	1	12
29	1	1	1	13	61	1	1	1	13
30	1	1	1	14	62	1	1	1	14
31	1	1	1	15	63	1	1	1	15
32	1	1	1	16	64	1	1	1	16
33	1	1	1	16	65	1	1	1	17
34	1	1	1	15	66	1	1	1	18
35	1	1	1	16	67	1	1	1	19;
36	1	1	1	2	end;				

Model:

```

param NPALETES;
param NPRATELEIRAS;
param NESTANTES;
param PALETE{i in 1..NPALETES,j in 1..4};
param PRATELEIRAS{i in 1..NPRATELEIRAS,j in 1..4};
param ESTANTES{i in 1..NESTANTES,j in 1..4};

var x{ i in 1..NPALETES, j in 1..NPRATELEIRAS, k in
1..NESTANTES},binary;

minimize tempototalacesso: sum {i in 1..NPALETES, j in
1..NPRATELEIRAS, k in
1..NESTANTES}PALETE[i,4]*PRATELEIRAS[j,4]*ESTANTES[k,4]*x[i,j,k];

s.t. arrumarpaletes{i in 1..NPALETES}:sum{j in 1..NPRATELEIRAS,k
in 1..NESTANTES}x[i,j,k]=1;

```

```

#fase 1: s.t. sobreposicao{j in 1..NPRATELEIRAS,k in
1..NESTANTES}:sum{i in 1..NPALETES}x[i,j,k]<=1;

#fase 2:
s.t. largura{j in 1..NPRATELEIRAS,k in 1..NESTANTES}: sum{i in
1..NPALETES}PALETE[i,3]*x[i,j,k]<=3.2;

solve;
display x;
display tempototalacesso;

end

```

Conclusão

Concluindo, tivemos dificuldades em realizar este projeto, devido ao facto deste problema não ser da nossa área e não termos grandes bases para resolver este problema para além do que foi aprendido ao longo deste semestre nesta unidade curricular, sendo que tudo aquilo que fizemos neste projeto foi feito graças ao que foi aprendido nas aulas.